School of Information Technology, Engineering, Mathematics and Physics

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CS241 – Software Design & Implementation
Semester 2, 2023
Mode: Face to Face

Short Test 2
Date: Tuesday 3 October 2023, Time: 11.00 – 11.50am (FJT)

Time Allowed: 50mins
30 marks (Weight: 10%)

Student ID: _____
Student Name: _____

LEARNING OUTCOMES:

- Develop software design document
- Develop a simple software system based on the design document

INSTRUCTIONS

1) This exam has three sections:
   a) Section A: Short-answer questions (25 marks)
   b) Section B: Programming (5 marks)
2) Answer all questions in this booklet. All questions are compulsory.
3) The total number of pages including this cover sheet is 6.

~ALL THE BEST~

Section A: Short Answer Questions *(25 Marks)*

1. Differentiate between Black box and glass box testing (2 marks).

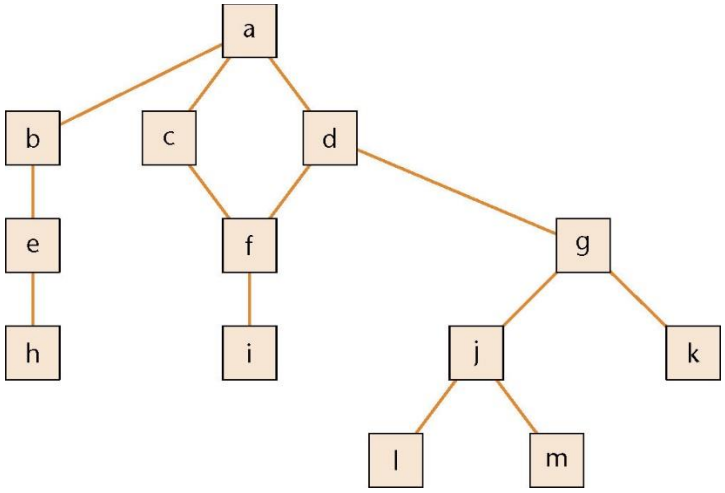
   Black box testing

   - Testing that ignores the internal mechanism of the system or component and focuses solely on the outputs in response to selected inputs and execution conditions.
   - Testing conducted to evaluate the compliance of a system or component with specified functional requirements.

   glass box testing

   Testing that takes into account the internal mechanism of a system or component.

2. Consider the following system with 13 modules and answer the questions as follows.



   a) List the way you can integrate this system using the top-down integration approach, i.e write the sequence of modules you will integrate using the top-down integration method. (2marks)

```
l,m,h,i,j,k,e,f,g,b,c,d,a OR
 h,e,b

i,f,c,d

l,m,j,k,g[d]

a [b,c,d]
```

b) Describe the top-down integration strategy and give one advantage of this method. (2 mark)

In the Top-Down Integration testing approach, the main module is designed at first then the submodules/subroutines are called from it. *Ensure an example is used to explain.

- Operational artifacts are thoroughly tested
- Operational artifacts are tested with drivers, not by fault shielding
- Fault isolation

- Major design faults are detected late

3. A program has three logical operations for type of sugar level are shown below:

Normal	Prediabetic	Diabetic
Less than 5	Between 5.1 and 8	More than 8.1

*Assume the argument "type" is of type sugar level and the method returns "X" for invalid level. Maximum Sugar level can be 20.

After implementing the method you are required to create unit test cases before your pull request can be processed. Using partition testing and equivalence partitioning method, which inputs will you pass in the test cases to cover all the valid and invalid partitions? (3 marks)

Valid: Sugar <= 5, 5.1 < Sugar <= 8, Sugar > 8.1,
Invalid: 0 < Sugar > 20

4. Difference between Verification and Validation in software testing. (2 marks)

"Verification – The process of evaluating a system or component to determine whether the products of a given development phase satisfy the conditions imposed at the start of that phase." "Are we building the product right". The software should conform to its specification.

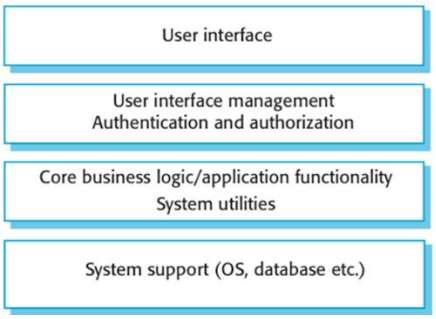
"Validation – The process of evaluating a system or component during or at the end of the development process to determine whether it satisfies specified requirements." "Are we building the right product". The software should do what the user really requires.

5. Find the errors in the following code and write in the provided space: (3 marks)

if (k < 2) {
  if (k > 3)
	for (j=0;j < 0;j++)
	  total = total + value[j];

x = x * k; }	
Infeasible Code (dead code) – ½ mark If (K>3) will never be executed – 1mark	Infeasible Code (dead code) – ½ mark For loop never executed -1mark

6. A generic layered architecture is shown below.



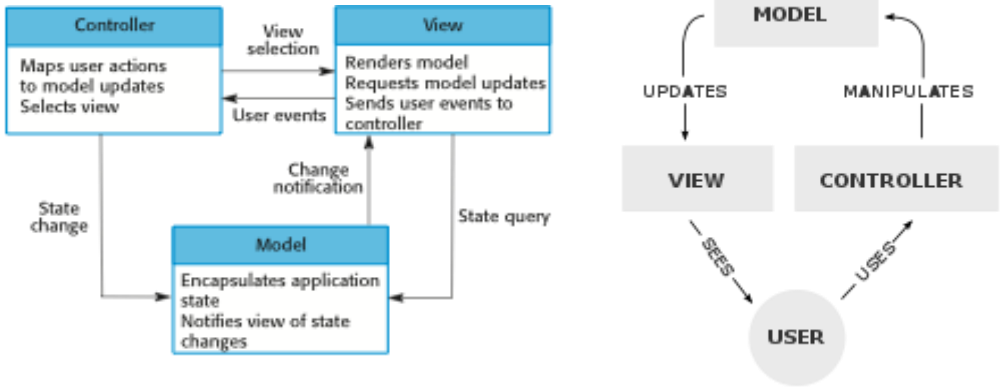
Using your knowledge learnt in CS241, describe layered architecture and give an advantage and disadvantage of this architecture. (3 marks)

Organises the system into a set of layers (or abstract machines) each of which provide a set of services.
Supports the incremental development of sub-systems in different layers. When a layer interface changes, only the adjacent layer is affected.

Adv: allows replacement of entire layers as long as interface is maintained. Redundundant facilities ed authentication can be provided in each layer to increase dependability of the system.
Dis: in practical providing clean separation between layers is often difficult and high-level layers may have to interact directly with lower-level layers rather than through the layer immediately below it.
Performance can be a problem

7. With help of the diagram describe the basic interactions in Model-View-Controller (MVC). (4 marks)

Model-View-Controller Pattern – Separates presentation and interaction from the system data. The Model component manages the system data and associated operations on that data. The View component defines and manages how the data is presented to the user. The Controller component manages user interaction passing the interactions to the View and the Model.



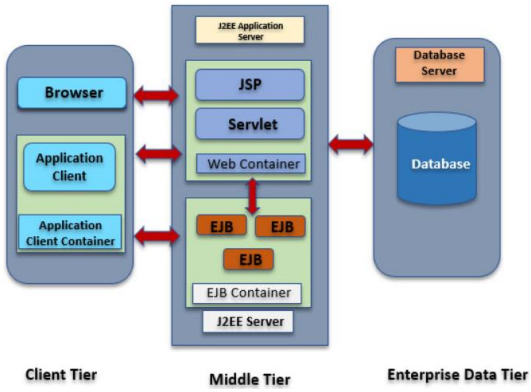
8. What is Software inspections and give one advantage of software inspection. (2 marks)

- Software inspections Concerned with analysis of the static system representation to discover problems (static verification). May be supplement by tool-based document and code analysis.
- These involve people examining the source representation with the aim of discovering anomalies and defects.
- Inspections not require execution of a system so may be used before implementation.

Adv:

During testing, errors can mask (hide) other errors. Because inspection is a static process, you don't have to be concerned with interactions between errors.
Incomplete versions of a system can be inspected without additional costs. If a program is incomplete, then you need to develop specialized test harnesses to test the parts that are available.
As well as searching for program defects, an inspection can also consider broader quality attributes of a program, such as compliance with standards, portability and maintainability.

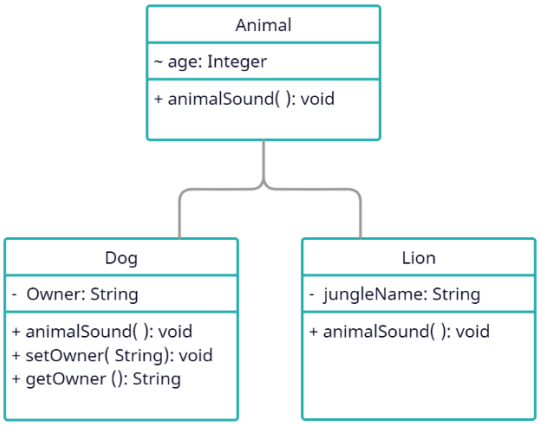
9. Shown below is the graphical representation of J2EE Architecture Tiers. What is the responsibility of client tier? (2 marks)



The Client tier is responsible for presenting data to the user, interacting with the user, and communicating with the other tiers of the application.
The Client tier is the only part the application the user ever sees.
The Client tier communicates with other tiers by way of well-defined interfaces.

Section B: Programming *(5 Marks)*

Use the following UML class diagram to answer the given questions:



a) Write the implementation of `Animal` class. Write any/all member functions implementation also. The `animalSound` should print "The animal makes a sound". (2 marks)

```
class Animal {
        protected int age;       1 mark

        public void animalSound() {
          System.out.println("The animal makes a sound"); 1 mark
        }
}
```

b) Write the implementation of `Dog` class. Write any/all member functions implementation also. The `animalSound` should print "`The Dog bark`". (3 marks)

```
class Dog extends Animal {
        private String Owner;     ½ mark
        public void animalSound() {
          System.out.println("The Dog bark");   1 mark
        }
        public void setOwner(String owner) {     1 mark
          Owner = owner;
```

```
        }
        public String getOwner(){      ½ mark
            return Owner;
        }
}
```

THE END

EXTRA ANSWER SHEET – USE ONLY WHEN NEEDED

- Mark the question number properly if used.